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Multi-Risk Dynamic Assessment for Resilience Building Towards Natural and Manmade Hazards of Submarine Infrastructures

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Submarine infrastructures are extremely important for the well-tempered function of the states and societies, but simultaneously less investigated at the level of resilience planning and more efficient protection towards threats and hazards. As an example of their importance, submarine cables are considered vital for global connectivity, as they are responsible for carrying 99% of international data traffic [1]. However, they face significant risks from natural disasters [2], accidental damage, and deliberate sabotage [3–5], especially in the emerging and liquid geopolitical environment of today. Recent incidents have highlighted the urgent need to enhance the security and resilience of these critical infrastructures to ensure their uninterrupted function, from the data transmission and the protection of the digital economy to the energy transmission and the protection of the energy needs. Proof of the ever-growing significance and attention that the states are putting on shielding submarine infrastructures towards this emerging landscape, is also the recommendations for 'Secure and Resilient Submarine Cable Infrastructures' [6] and the 'Joint Communication to strengthen the security and resilience of submarine cables' [7] that the EU has published recently (2024 and 2025 respectively).

The importance of building resilience in today's world is highlighted importantly the recent years and actions have been taken towards this direction, both from the technical/research community and the states, but yet it's not fully adopted in practice as main approach to various problems regarding critical infrastructures and the landscape of current hazards and threats. Building resilience to natural, climatic and man-made events must be of prime importance in an ever-growing and interconnected world, including especially any type of critical infrastructures, meaning also the submarine. Not only for safety and security reasons, but also regarding the economic impact that the absence of resilient planning and philosophy can impose on the states and the operators, as the cost of not being resilient is important. According to respective studies [8] which are based on over 3,000 scenarios, the urgency of designing resilient infrastructure is clear when considering natural hazards: in most scenarios, it is costly to delay global action on resilience to 2030, with a median value of US$ 1.0 trillion in potential losses. Resilience sits in an intricate interplay among individuals, communities, institutions and infrastructures. Moreover, it is well assessed that resilience depends on many factors such as technological (e.g. platforms and tools), human (e.g. capability of intervention and exploitation of information from citizens and users) and user's acceptance (understanding the added value provided by novel technological solutions for the improvement of operative capabilities, in ordinary and extraordinary conditions).

The aim of the research is to provide a systematic methodological approach to assess the resilience capacity of the European critical infrastructure sectors of submarine telecommunication cables, power cables and gas pipelines. Based on the multi-risk dynamic assessment of natural and manmade hazards which is successfully tested also on other types of infrastructures (e.g., gas pipelines) and with the respective adjustments for this type of critical infrastructures, the operators will have the situational overview in order to further strengthen their assets and make then more resilient and robust. The novelty of this approach is based on its proven efficiency to various other types of infrastructure assets and its adjustment for the needs of the submarine infrastructures, a sector of infrastructure assets which is considered less investigated by the research community regarding its critical role and therefore vulnerable. The research conducted is part of the EU-funded research project of VIGIMARE.

Introduction

Submarine infrastructures are extremely important for the well-tempered function of the states and societies, but simultaneously less investigated at the level of resilience planning and more efficient protection towards threats and hazards. As an example of their importance, submarine cables are considered vital for global connectivity, as they are responsible for carrying 99% of international data traffic [1]. However, they face significant risks from natural disasters [2], accidental damage, and deliberate sabotage [3–5], especially in the emerging and liquid geopolitical environment of today. Recent incidents have highlighted the urgent need to enhance the security and resilience of these critical infrastructures to ensure their uninterrupted function, from the data transmission and the protection of the digital economy to the energy transmission and the protection of the energy needs. Proof of the ever-growing significance and attention that the states are putting on shielding submarine infrastructures towards this emerging landscape, is also the recommendations for 'Secure and Resilient Submarine Cable Infrastructures' [6] and the 'Joint Communication to strengthen the security and resilience of submarine cables' [7] that the EU has published recently (2024 and 2025 respectively).

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Methodology

The Multi-risk Dynamic Assessment (MrDA) general methodology which is adopted aims to offer an accurate and precise resilience assessment of submarine infrastructure assets to the respective industrial stakeholders, policymakers, authorities and infrastructure operators. It provides a crucial depiction of the assets' condition as it at these users, in order to proactively prepare for and effectively respond to disruptive events, including natural disasters, cyberattacks, equipment failures and hybrid threats. MrDA has five steps (Fig. 1), and are described epigrammatically below:

- System Characterization, where the description of the entire system and relevant assets to be analyze are included (e.g., the location of the system in the external context; identification of the system's elements/assets; identification of relationships between the system's elements/assets)
- Asset Characterization: where the description of the individual asset and its qualifying (technical) characteristics necessary for the analysis will be collected (e.g., dimensions, mechanical parameters)
- Hazard Characterization: where takes place the evaluation of possible disruptive events divided (based on its type into) climate and natural hazards (e.g., earthquake, wind, flood), technological and man-made hazards (e.g., cyber-threats, sabotage), followed by the definition of relevant hazard scenarios and relevant hazard modeling in the area and assets considered (based on technical parameters like frequency and magnitude)
- Risk, Vulnerability, Consequences and Impact analysis, where starting from the defined assets and the hazards considered, they are conducted analysis of the assets' risks, vulnerability consequences and the quantification of the impacts
- Dynamic risk assessment, where respective analysis carried out on the asset(s) considered and/or on the entire system considering cascading effects, exploiting the restoration curves for the consequences and deterministic/probabilistic disruption functions for the disruption analysis, with the results to be expressed in term of avoided losses (after being calculated the expected annual losses)

The methodology is foreseen to be scalable and adaptable to each component of the submarine infrastructure asset under examination. The level of detail of each step can be simplified or complicated based on the different case studies to be studied and the different conditions. This adaptability ensures that the methodology can be applied to a wide range of situations, from high-level, strategic assessments to detailed analyses of specific equipment or systems. For instance, when analyzing a subsea pipeline, the steps might involve aggregating data and focusing on interdependencies between major joints. This flexibility makes the methodology a valuable tool for various stakeholders, allowing them to tailor the assessment to their specific needs and resources.

The overall goal is to minimize both direct and indirect losses, reduce the duration and severity of service disruptions, and enhance the overall resilience of submarine infrastructures. The methodology will offer procedures for identifying potential vulnerabilities, assessing the cascading effects of disruptions, and implementing mitigation strategies.

Figure 1—Schematic representation of the five-step MrDA

System and Asset Characterization

The first step of the methodology involves a comprehensive description of the submarine infrastructure. This includes detailing all the properties inherent to a system network, such as its topology, capacity, connectivity, and operational characteristics. Furthermore, it encompasses the precise localization and identification of key assets, ranging from power generation units and cable substations to transmission lines and critical control systems, along with a thorough understanding of their inter-dependencies and how they rely on each other for seamless operation.

For every critical asset or component of the system, gathering mechanical and geometrical data (along with economic, environmental, organizational) is essential for the asset characterization. The precision and level of detail in this data directly dictate the accuracy of the analysis for each asset. Consequently, a comprehensive dataset enables a more thorough and nuanced assessment. For each main step of the MrDA, specific data are needed (e.g., location and intensity for hazard curves, physical and geometrical details for vulnerability, data on affected populations).

Hazard Characterization

The initial step in the hazard analysis involves examining the network's geographical location to identify the most probable natural and manmade hazards. This requires researching historical occurrences and consulting hazard maps from government agencies or scientific institutions. The MrDA considers various types of natural hazards, including geophysical (Earthquakes), hydrological (Flooding, Landslides, and Debris flow), and climatological (Heavy snowfall and extreme weather) events and manmade ones, including sabotage and terrorist attacks. Usually, these hazards are measured in terms of intensity (Intensity Measure = IM).

The level of detail in the hazard information significantly impacts subsequent analyses. Ideally, with complete knowledge of the hazard, meaning all possible events are identified, a hazard curve (Figure 2) can often be generated. Otherwise, simplified hazard maps can be utilized without full knowledge of the possible intensity measures for all the return periods. In this case, the few data available (IMs) can be used to build up the hazard curve through a fitting process through power or exponential laws. An example of this approach is that in case of earthquake or wind and extreme events where several return periods are available in national codes or websites database (Figure 8). This figure displays two hazard curves. One curve illustrates the relationship between the peak ground acceleration (a measure of earthquake intensity) and its mean annual frequency (the average number of times that level of acceleration is expected to be exceeded in a year). The other curve shows the relationship between the peak velocity (a measure of flood intensity) and its corresponding mean annual frequency. If a hazard curve is not easily obtainable, a dedicated model must be defined.

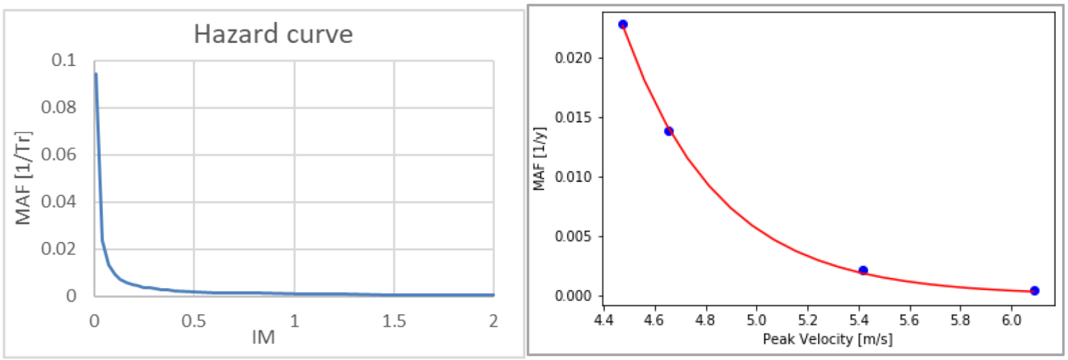

Figure 2—Hazard curve for Earthquake and Wind

Vulnerability, Consequences and Impact Analysis

Vulnerability analysis. Vulnerability of submarine infrastructures to natural and manmade hazards refer to the susceptibility of the various components and the overall functioning of the relevant systems to damage, disruption, or failure when exposed to natural events and manmade threats. It essentially quantifies how much a relevant asset or the entire system is likely to be negatively affected by a natural hazard of a given intensity (IM). This is determined by a combination of factors related mainly to the asset itself and the characteristics of the hazard.

This assessment can be tailored by employing various analysis methods, depending on the specific needs. It can be performed as a high-level assessment (through indicators or matrices) or through a more in-depth analysis considering specific weaknesses of the asset against natural events (e.g. fragility curves). Fragility curves are essential to understanding asset vulnerability as they provide the probability of exceeding a specific damage level for a given intensity measure. For each asset, four damage levels are defined, corresponding to four criticality levels. For each damage state, there are specified[1)] the percentage of damage, specific to each level and asset (recorded in an internal database) and[2)] a description of the observable impact on the facility for each damage level. This subdivision allows for the assignment of varying criticality levels to different subsectors based on the specific asset being considered.

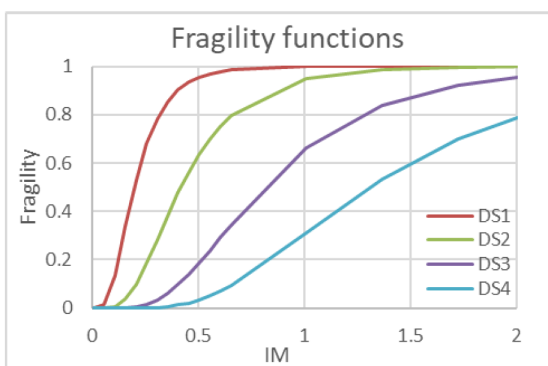

Figure 3—Fragility curve example

Consequence and disruption analysis. Consequence analysis allows for the estimation of the loss of functionality of critical assets across various damage states. While direct losses to assets from disruptive events are calculated based on the level of damage and the associated economic loss, indirect losses require knowing the asset's operational levels. Therefore, a reduction in the operativity or functionality of infrastructure assets due to disruptive events leads to indirect economic losses for the surrounding affected community. By combining information from consequence and disruption functions and fragility functions for each limit state, it can be defined an estimation of the economic loss. Fragility functions predict the likelihood (probability) of an asset reaching a specific damage state given an event intensity measure. Consequence and disruption functions describe the extent of damage, the downtime, and the recovery of an asset's functionality over time after reaching a particular damage state.

Economic impact assessment. Impact concerns the presence of individuals, livelihoods, species or ecosystems, environmental functions, services, resources, infrastructure, or economic, social, and cultural assets in places and contexts where a hazard could have detrimental consequences. The impact is normally quantified in the determination of damages and losses caused to stakeholders, environment and human life. The impact analysis is structured around three distinct categories:

- Impacts on People (fatalities/injuries)
- Impacts on the Physical System/Infrastructure (structural damages)
- Impacts on Service Continuity (interruptions/downtime)

Each component's evaluation is also expressed in economic terms, allowing for their combination to yield a single, unified impact value, to be used in the risk assessment.

Regarding the impact on people, the impact is carried out determining the number of fatalities [9] and severe injuries caused by an event.

$$Death_{DSi} = Death_{DSi}(\%) \cdot \delta_t \cdot People\_in\_asset$$
$$Severe\_njured_{DSi} = Severe\_njured_{DSi}(\%) \cdot \delta_t \cdot People\_in\_asset$$

where $Death_{DSi\,(\%)}$ and $Severe\ injured_{DSi}\ (\%)$ are percentage of people related to the damage state considered, and δ_t is a coefficient that counts the people evacuation capacity the number of people in the asset if present.

Regarding the physical impact it can be determined as:

$$ID_{DSi} = Damage_{DSi}(\%) \cdot Reconstruction\_Cost$$

where $Damage_{DSi}$ (%) is the level of damage for each damage state and *Reconstruction_Cost* is the expected cost to take back the asset to full operation.

This last parameter can be calculated in two different ways:

- Directly, using the construction cost derived from the asset datasheet. If available, the construction cost must be adjusted to current values considering the year of construction and inflation rates.
- Indirectly, using dimensions derived from the asset datasheet and unitary costs obtained from literature. The indirect economic losses can be related to a disruption event and to the number of people affected by the assets' functions (e.g., loss of internet for their businesses) after the hazard.

The total impact ($I_{TOT,DSi}$) is evaluated by summing the different economic losses categories per each damage state, as shown in the following formula:

$$I_{TOT,DSi} = I_{P\_DSi} + I_{D\_DSi} + I_{R\_DSi}$$

where: I_{P_DSi} is the Impact on People economic loss; I_{D_DSi} is the Impact on Physical System economic loss; I_{R_DSi} is the Impact on Service Continuity economic loss.

Dynamic Risk Assessment

A complete understanding of Multi-risk Dynamic Assessment necessitates the effective combination of Hazard, Vulnerability, and Impact. The initial crucial step involves defining both Hazard and Vulnerability and subsequently integrating their functions to derive the probability of damage (Figure 4). The Damage probability per each damage state can be computed as the convolution integral between the Hazard curve, if available, and the fragility curves of the limit state DS.

$$P_{D,DSi} = \int_0^{+\infty} F_{R,DSi}(IM) * H_S(IM) d(IM)$$

where FR represents the fragility curve and HS the hazard curve, both expressed as a function of IM

Following this, by combining the calculated probability of damage with the potential economic impacts, the final risk assessment can be determined, offering a comprehensive evaluation of potential losses and their likelihood. The proposed methodology quantifies dynamic physical risk using the Expected Annual Loss (EAL). The EAL, which is considered the area under the final Loss curve of Figure 5, represents the anticipated average loss per year, expressed as a percentage of the asset's reconstruction value. This key metric synthesizes various risk assessment components, specifically the likelihood of damage (determined by Hazard and Vulnerability) and the overall consequences (shaped by economic impacts).

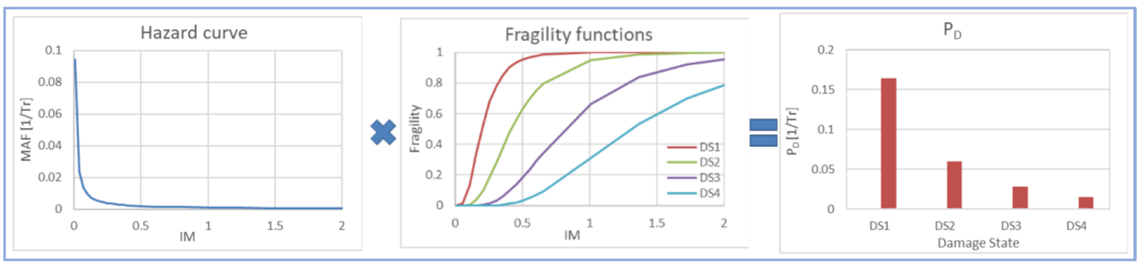

Figure 4—Probability of damage assessment

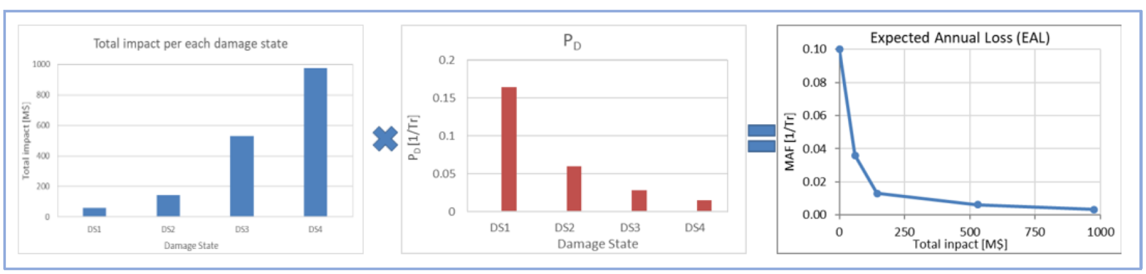

Figure 5—EAL assessment

In essence, EAL provides a forward-looking perspective on risk. It's not just a snapshot of current conditions but rather a projection of the average loss expected in a year, considering the interplay of hazards, vulnerabilities and impacts. This makes it invaluable knowledge for:

- **Prioritization**: Comparing the EAL of different risks allows for informed decisions about where to allocate resources for risk reduction and mitigation efforts. Higher EAL values typically warrant more immediate attention.
- **Cost-Benefit Analysis**: When evaluating potential risk mitigation strategies, the reduction in EAL achieved by the strategy can be weighed against the cost of implementing it.
- **Performance Monitoring**: Tracking changes in EAL over time can indicate the effectiveness of risk management strategies and highlight emerging risks.
- **Communication**: A single EAL figure can be a powerful way to communicate the level of risk to stakeholders in a clear and understandable manner.

Therefore, EAL serves as a cornerstone in this dynamic risk assessment methodology by providing a quantifiable and comprehensive measure that integrates the various facets of risk and allows for informed decision-making in the face of uncertainty.

Results and Conclusions

The above-mentioned MrDA aims to support the overall technical solutions which are being developed in VIGIMARE and are having as a goal to build a near real-time situational awareness picture onshore and offshore above the cables and pipelines at critical areas and intersections, with an automated anomaly recognition service [10]. To do so, and in order to improve situational awareness both onshore and offshore, it aims to develop sensing and analysis services, as well as a Virtual Control Room. Within the project, the plan is to stress test the solution with data from real incidents and validate it in three different European sea areas (Mediterranean Sea, Irish Sea and Baltic Sea), in order to promote its valuable outcome to the European society [10]. The MrDA therefore is integral part of depicting the current resilient conditions of the assets as they are and feeding with relevant information the rest of the tools, while it is being implemented at assets ranging to three different seas and exploits also data and knowledge from real incidents.

At this point, the MrDA is being adjusted to the special needs and characteristics of the submarine infrastructures, in complete alignment and under a common framework with cyber-threats risk assessment. Some natural hazards which are becoming crucial for these types of assets will be highly investigated (e.g., tsunamis) and the surrounding geopolitical context along with other manmade threats are part of the assessment. The outer goal of the multi-risk analysis is the protection of the assets towards both natural and manmade threats and hazards, in the most effective way, based on the knowledge of previously succeeded EU-funded projects. The initial results for the adjustment of the methodologies designed from other types of assets to submarine ones are promising.

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